Screening of Scalar-Field Clock Couplings: Cavity–Atom Gauge Resolution, BACON Constraints, and Nuclear Clock Predictions in Density Field Dynamics

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Abstract

In Density Field Dynamics (DFD), the scalar refractive field ψ couples to Standard Model sectors through the optical metric, producing species-dependent clock frequency shifts. We resolve a gauge ambiguity in the photon sector coupling by proving that the optical metric's constitutive relations ($\varepsilon = \varepsilon_0 e^{\psi}$, $\mu = \mu_0 e^{\psi}$) enforce geometric cancellation of cavity and atomic responses at tree level. The physical LPI violation is a one-loop residual, screened by the local gravitational environment.

Three independent empirical checks confirm the cancellation: (i) fine-structure splitting constraints ($> 10^7$), (ii) PTB Yb^+ E3/E2 same-ion comparison ($\sim 100\times$), and (iii) BACON three-species optical clock network ($\sim 10^6$). The PTB E3/E2 bound [7] further constrains pure electromagnetic-sector coupling to $|k_\alpha| < 1.4 \times 10^{-9}$, confirming that the dominant clock coupling is composition-dependent and strong-sector, not purely α-dependent. The BACON data [2] constrain the screening regime: solar-orbit screening ($k_\alpha^{\rm eff} \approx 2.4 \times 10^{-5}$) is excluded by $\sim 100\times$, while Earth-surface screening ($k_\alpha^{\rm eff} \approx 6 \times 10^{-7}$) is consistent with the observed between-day variability.

We derive predictions for the ^{229}Th nuclear clock transition, whose recently measured α-sensitivity $K = 5900 \pm 2300$ [4] amplifies even small couplings to detectable levels. The primary prediction comes from the strong-sector coupling ($k_s^{\rm eff} \sim 2.4 \times 10^{-6}$, $S_{\rm Th}^{\alpha_s} \sim 10^4$ [8]), giving an annual modulation $\delta R/R \sim 4 \times 10^{-12}$ in the Th-229/Sr ratio—within reach of current nuclear clock programs. A composition-dependent lower bound from the family model gives $\sim 10^{-14}$. General relativity predicts exactly zero.

1 Introduction

Density Field Dynamics (DFD) replaces general relativity's curved spacetime with a scalar refractive field ψ on flat $\mathbb{R}^3$, where light and matter respond to ψ gradients as in an optical medium [1]. The optical metric

$$d\tilde{s}^2 = -\frac{c^2}{n^2}dt^2 + d\mathbf{x}^2, \qquad n = e^{\psi}, \tag{1}$$

reproduces all solar system tests of GR (PPN: $\gamma = \beta = 1$), while predicting species-dependent clock couplings that constitute the theory's sharpest experimental signature.

A key prediction concerns the cavity–atom comparison: a test comparing optical cavity resonances with atomic clock frequencies at different gravitational potentials. An earlier treatment (Sec. 10 of DFD v3.1) assigned independent coupling constants $K_\gamma \approx 1$ (photon sector) and $K_{\rm atom} \approx 0$ (atomic sector), predicting an LPI violation parameter $\xi_{\rm LPI} \approx 1$–2.

This paper demonstrates that this assignment contains a gauge error. The optical metric (1) uniquely determines vacuum constitutive relations that modify the Coulomb interaction, causing electromagnetic cavity spacers to expand in exactly the way needed to compensate the light-speed reduction. The tree-level cancellation reduces $\xi_{\rm LPI}$ by a factor $\sim 10^5$, with the physical residual being a screened one-loop quantum correction.

We present three independent empirical confirmations, analyze the BACON optical clock network data to constrain the screening regime, and derive predictions for the ^{229}Th nuclear clock. The coupling channel structure—clarified by the PTB Yb^+ E3/E2 constraint [7]—demonstrates that the strong sector,

not the electromagnetic sector, carries the dominant signal for nuclear clock transitions. This makes Th-229 uniquely powerful: atomic clocks are blind to the channel that nuclear clocks access.

2 The Constitutive Chain

2.1 Optical metric determines permittivity

The Tamm–Plebanski formalism maps any effective spacetime metric to vacuum constitutive relations for electromagnetism. For the DFD optical metric (1), with $g_{00} = -c^2 e^{-2\psi}$ and $g_{ij} = \delta_{ij}$, the standard construction yields:

$$\varepsilon_{\text{eff}} = \varepsilon_0\, e^{+\psi}, \qquad \mu_{\text{eff}} = \mu_0\, e^{+\psi}. \tag{2}$$

This is an impedance-matched medium ($Z = Z_0$) with refractive index $n = c\sqrt{\varepsilon_{\text{eff}}\mu_{\text{eff}}} = e^{\psi}$.

The constitutive relations (2) are not a modeling choice; they are uniquely determined by the metric. Any departure would violate the causal structure encoded in (1).

2.2 Coulomb potential and fine-structure constant

Virtual photons—mediating the static Coulomb interaction—propagate on the same optical background. The electrostatic potential between charges separated by coordinate distance r is:

$$V(r) = \frac{e^2}{4\pi\varepsilon_{\text{eff}}\, r} = V_0(r) \times e^{-\psi}. \tag{3}$$

The Coulomb force weakens in stronger gravitational fields.

The fine-structure constant, defined as a dimensionless ratio measured locally:

$$\alpha(\psi) = \frac{e^2}{4\pi\varepsilon_{\text{eff}}\,\hbar\, c_{\text{local}}} = \frac{e^2}{4\pi(\varepsilon_0 e^{\psi})\hbar(c\, e^{-\psi})} = \alpha_0. \tag{4}$$

The $e^{-\psi}$ from enhanced permittivity and the $e^{+\psi}$ from reduced light speed cancel exactly. Therefore, **α is ψ-independent at tree level**.

This is not accidental. It is the electromagnetic manifestation of the Weak Equivalence Principle (WEP): in a local freely-falling frame, electromagnetic coupling constants are position-independent. DFD satisfies WEP by construction, since its PPN parameters match GR exactly.

2.3 Atomic frequency scaling

With α constant, the Bohr radius scales as:

$$a_0(\psi) = \frac{\hbar}{m_e\, c_{\text{local}}\, \alpha} = a_0^{(0)}\, e^{+\psi}. \tag{5}$$

The Rydberg energy scales as $E_R \propto m_e c_{\text{local}}^2 \alpha^2 \propto e^{-2\psi}$. For a transition with relativistic correction parameter ϵ_A (encoding the sensitivity to α beyond leading order):

$$f_A \propto e^{-(2+\epsilon_A)\psi}, \tag{6}$$

where ϵ_A is related to the standard sensitivity coefficient by $S_A^{\alpha} = 2 + \epsilon_A$ for optical transitions. For common transitions: $\epsilon_{\text{Sr}} = 0.06$, $\epsilon_{\text{Yb}} = 0.31$, $\epsilon_{\text{Al}^+} = 0.008$.

2.4 Cavity frequency scaling

A Fabry–Pérot cavity with electromagnetic spacer (silicon, ULE glass) has lattice constant $\propto a_0(\psi) \propto e^{+\psi}$ and thus spacer length $L \propto e^{+\psi}$. The resonance frequency is:

$$f_{\text{cav}} = \frac{m\, c_{\text{local}}}{2\, L(\psi)} \propto \frac{e^{-\psi}}{e^{+\psi}} = e^{-2\psi}. \tag{7}$$

The two effects — slower light ($e^{-\psi}$) and longer spacer ($e^{+\psi}$) — compound to give $e^{-2\psi}$, identical to the leading atomic scaling.

2.5 The cancellation

The cavity–atom ratio:

$$R = \frac{f_{\text{cav}}}{f_A} \propto \frac{e^{-2\psi}}{e^{-(2+\epsilon_A)\psi}} = e^{+\epsilon_A\,\psi}. \tag{8}$$

The universal $e^{-2\psi}$ gravitational redshift cancels. The geometric residual $\xi_{\text{geom}} = \epsilon_A$ (e.g., 0.06 for Sr/Si) is an artifact of coordinate-time expressions. In proper time, $\alpha = \alpha_0$ exactly (Eq. 4), so R is constant.

This resolves the gauge ambiguity. The earlier treatment incorrectly assigned $K_\gamma = 1$ by accounting for light-speed reduction ($c_{\text{local}} = c\, e^{-\psi}$) without simultaneously accounting for the spacer expansion forced by the same constitutive relations. Including both effects, the tree-level signal vanishes.

3 The Physical Residual: Screened Coupling

The tree-level WEP protection is broken at one loop by Unruh–de Sitter screening of quantum vacuum fluctuations. In DFD, an atom at local gravitational acceleration a is bathed in Unruh radiation at temperature $T_U = \hbar a/(2\pi k_B c)$. When $T_U \gg T_{dS}$ (the de Sitter temperature associated with the cosmological horizon), the Unruh bath screens the atom's sensitivity to the cosmological vacuum state.

The screening produces an effective coupling:

$$k_\alpha^{\rm eff}(a) = 2\sqrt{\alpha}\,\mu_{\rm LPI}\left(\frac{a}{a_0}\right), \tag{9}$$

where the LPI interpolation function is:

$$\mu_{\rm LPI}(y) = \frac{1}{\sqrt{1+y}}, \qquad y = a/a_0, \tag{10}$$

and $a_0 \approx 1.2\times10^{-10}$ m/s^2 is the MOND acceleration scale (derived, not fitted, in DFD).

For *cross-species* clock comparisons, the measurable LPI violation depends on the full coupling structure (see Sec. 6). The effective coupling Eq. (9) sets the overall scale of the one-loop residual; the channel through which it operates (electromagnetic, strong, or compositional) is determined by the comparison type.

4 Three Independent Confirmations

4.1 Check 1: Fine-structure splitting ratio

If α varied geometrically as $\alpha_0 e^{-\psi}$ (unscreened), the ratio of two transitions in the same atom with different α-sensitivities would show annual modulation at amplitude:

$$\frac{\delta R}{R} = \Delta S^\alpha \times \delta\psi_{\rm annual} \approx \Delta S^\alpha \times 1.65\times10^{-10}. \tag{11}$$

For any pair with $|\Delta S^\alpha| \sim 1$, this gives $\sim 10^{-10}$. Precision spectroscopy constrains annual variation in transition ratios to $< 10^{-17}$.

Ruled out by $> 10^7$.

4.2 Check 2: PTB Yb$^+$ E3/E2

Lange et al. [7] compare the E2 ($S^\alpha = +1.0$) and E3 ($S^\alpha = -5.95$) transitions in a single trapped ^{171}Yb$^+$ ion. Their measured annual modulation amplitude is $(-16\pm13)\times10^{-18}$, consistent with zero, constraining $|K_{E3} - K_{E2}| < 10^{-8}$.

Geometric prediction: $|\Delta S^\alpha| \times \delta\psi_{\rm annual} = 6.95 \times 1.65\times10^{-10} = 1.15\times10^{-9}$.

Ruled out by $\sim 100\times$.

This result has a deeper implication beyond confirming the geometric cancellation. Since E3 and E2 are transitions in the *same ion*, they share identical nuclear composition ($\Delta C_N = 0$), identical electronic structure ($\Delta C_e = 0$), and zero strong-sector sensitivity ($\Delta\kappa_s = 0$). Only the α-sensitivity differs ($\Delta S^\alpha = -6.95$). The E3/E2 comparison therefore isolates the pure α-sector, giving:

$$|k_\alpha| < \frac{10^{-8}}{6.95} = 1.4\times10^{-9}. \tag{12}$$

This is consistent with DFD's composition-dependent coupling model: same-ion comparisons cancel the composition terms that carry the cross-species signal, leaving only the negligible α-sector residual.

4.3 Check 3: BACON three-species network

Beloy et al. [2] measure frequency ratios of Al$^+$, ^{87}Sr, and ^{171}Yb at uncertainties of 6–8 $\times 10^{-18}$ over 8 months (Nov 2017–Jun 2018) spanning perihelion.

The α-sensitivity coefficients used in that work are: $K_{\rm Al^+} = 0.008$, $K_{\rm Sr} = 0.06$, $K_{\rm Yb} = 0.31$.

For Yb/Sr, $\Delta K = 0.25$. Geometric prediction: $0.25 \times 1.65\times10^{-10} = 4.1\times10^{-11}$. Observed Yb/Sr stability: WSD $= 1.1\times10^{-17}$.

Ruled out by $\sim 4\times10^6$.

The dark matter search in Fig. 4 of that paper includes the annual frequency ($f_{\rm orbit} \approx 3.17\times10^{-8}$ Hz) in its search band, unlike Kennedy et al. [3], whose 12-day dataset excludes annual frequencies entirely.

5 BACON Screening Regime Constraint

The BACON data distinguish between two screening scenarios.

5.1 Solar-orbit screening

If screening is determined by the gravitational acceleration at Earth's orbital radius ($a = 5.93 \times$

10^{-3} m/s^2, $y = 4.94 \times 10^7$):

$$k_\alpha^{\text{eff}} = 2\sqrt{\alpha} \times (4.94 \times 10^7)^{-1/2} = 2.43 \times 10^{-5}. \quad (13)$$

Predicted Yb/Sr annual signal: $0.25 \times 2.43 \times 10^{-5} \times 1.65 \times 10^{-10} = 1.0 \times 10^{-15}$.

The BACON Yb/Sr WSD is 1.1×10^{-17}.

Excluded by $\sim 90\times$.

5.2 Earth-surface screening

If screening is determined by the local field at Earth's surface ($a = 9.8$ m/s^2, $y = 8.17 \times 10^{10}$):

$$k_\alpha^{\text{eff}} = 2\sqrt{\alpha} \times (8.17 \times 10^{10})^{-1/2} = 5.98 \times 10^{-7}. \quad (14)$$

Predicted Yb/Sr annual signal: $0.25 \times 5.98 \times 10^{-7} \times 1.65 \times 10^{-10} = 2.47 \times 10^{-17}$.

The BACON between-day variability for Yb/Sr is $\xi = 10.8 \times 10^{-18}$ with $\chi^2_{\text{red}} = 6.0$—significant excess scatter whose source was not identified. The predicted signal is $\sim 2.3\times$ the observed between-day variability.

Consistent with data (marginally above scatter, but only 9 measurement days with poor annual-phase coverage).

Caveat. The BACON campaign comprised ~ 17 measurement days over ~ 8 months, providing limited phase coverage of the annual cycle. The between-day variability constrains the annual modulation amplitude only indirectly; a dedicated year-long campaign with continuous monitoring would provide a sharper test. The screening regime conclusion—that solar-orbit screening is excluded while Earth-surface screening is consistent—is robust, but the quantitative comparison to BACON scatter should be interpreted with caution.

5.3 Physical interpretation

The screening must be evaluated at the *local* gravitational environment. Clocks on Earth's surface sit at acceleration $a = 9.8$ m/s^2, dominated by Earth's own field. The solar annual variation ($\delta\psi \sim 10^{-10}$) is a small perturbation on top of a much more strongly screened background.

This is physically consistent with the Unruh–de Sitter mechanism: the thermal bath that screens quantum vacuum fluctuations depends on the *total* local acceleration, not on the source of any particular perturbation.

6 Coupling Channel Structure

The PTB E3/E2 constraint (Sec. 4.2) has implications beyond confirming the geometric cancellation. It reveals which coupling channels can carry a detectable signal.

6.1 General coupling formula

DFD predicts that the gravitational coupling coefficient for clock species A has contributions from multiple sectors [1]:

$$K_A = k_\alpha \kappa_\alpha^{(A)} + k_s \kappa_s^{(A)} + k_N C_N^{(A)} + k_e C_e^{(A)}, \quad (15)$$

where $\kappa_\alpha^{(A)} = S_A^\alpha$ and $\kappa_s^{(A)} = S_A^{\alpha_s}$ are the electromagnetic and strong-sector sensitivities, $C_N^{(A)}$ and $C_e^{(A)}$ are nuclear and electronic family charges (Appendix T of [1]), and k_α, k_s, k_N, k_e are coupling constants.

6.2 What E3/E2 constrains and what it does not

The same-ion E3/E2 comparison has $\Delta C_N = \Delta C_e = \Delta\kappa_s = 0$, isolating the α-sector: $\Delta K_{\text{E3/E2}} = k_\alpha \times (-6.95)$. The PTB bound therefore constrains:

$$|k_\alpha| < 1.4 \times 10^{-9}. \quad (16)$$

This eliminates the pure α-sector as a significant coupling channel. However, it leaves *unconstrained*:

1. The strong-sector coupling k_s, since atomic transitions have $\Delta\kappa_s = 0$.
2. The composition-dependent couplings k_N, k_e, since same-ion comparisons have $\Delta C_N = \Delta C_e = 0$.

6.3 Effective parameterization for cross-species comparisons

For cross-species *atomic* clock comparisons (e.g., Yb$^+$/Sr, Cs/Sr), the α-sector is negligible by Eq. (12), and the strong-sector sensitivities κ_s are zero (electronic transitions). The signal comes entirely from composition differences:

$$\Delta K_{AB} = k_N \Delta C_N + k_e \Delta C_e. \quad (17)$$

An important empirical observation is that the family charges C_N, C_e correlate with the α-sensitivities S_A^α for the atomic species commonly used

in precision clock comparisons. Both quantities track relativistic corrections that scale with nuclear charge Z. This means the simplified formula $K_A \approx k_{\text{eff}} \cdot S_A^\alpha$ works as an empirical approximation for cross-species atomic comparisons, even though the physical coupling is through composition, not through α directly.

This correlation explains why the DFD framework successfully describes the pattern of hints and nulls in existing clock data: cross-species pairs with large $|\Delta S^\alpha|$ tend to also have large $|\Delta C_N|$, producing real signals; same-ion pairs (E3/E2) have zero ΔC_N regardless of ΔS^α, producing nulls.

6.4 Nuclear clock comparisons: the strong sector

The ^{229}Th nuclear isomer transition is qualitatively different because the 8.4 eV energy arises from near-cancellation between Coulomb ($\sim +300$ keV) and nuclear strong-force (~ -300 keV) contributions. This makes the transition sensitive to α_s through dimensional transmutation [8], with $S_{\text{Th}}^{\alpha_s} \sim \mathcal{O}(10^4)$.

DFD predicts that each gauge sector couples to ψ with strength $k_i = \alpha_i^2/(2\pi)$ [1]. For QCD at the confinement scale ($\alpha_s \sim 0.3$–0.5):

$$k_s = \frac{\alpha_s^2}{2\pi} \approx 2.2 \times 10^{-3}. \tag{18}$$

The ratio $k_s/k_\alpha \approx 260$ means that nuclear transitions couple $\sim 260\times$ more strongly to ψ per unit sensitivity than electronic transitions. Combined with the dimensional transmutation amplification ($\delta\Lambda_{\text{QCD}}/\Lambda_{\text{QCD}} \approx 64\,\delta\alpha_s/\alpha_s$), the nuclear clock accesses a coupling channel that is entirely invisible to atomic clocks—and entirely unconstrained by the PTB E3/E2 bound.

7 Nuclear Clock Predictions

7.1 Strong-sector: the primary prediction

With Earth-surface screening applied to the strong sector by analogy with the electromagnetic sector:

$$k_s^{\text{eff}}(a_\oplus) = 2\sqrt{\alpha_s}\,\mu_{\text{LPI}}(y_\oplus) \approx 2.4 \times 10^{-6}, \tag{19}$$

where $\alpha_s \approx 0.118$ at M_Z (the relevant scale for the vertex-counting argument).

Using Flambaum's estimate [8] $S_{\text{Th}}^{\alpha_s} \sim 10^4$:

$$\begin{aligned}\left(\frac{\delta R}{R}\right)_s &= k_s^{\text{eff}} \times S_{\text{Th}}^{\alpha_s} \times \delta\psi_{\text{annual}} \\ &= 2.4 \times 10^{-6} \times 10^4 \times 1.65 \times 10^{-10} \\ &= \mathbf{4.0 \times 10^{-12}}.\end{aligned} \tag{20}$$

On line b at 195 K: $\delta\nu_b \sim 8$ kHz (half-amplitude).

Uncertainty. Both k_s^{eff} and $S_{\text{Th}}^{\alpha_s}$ carry order-of-magnitude uncertainty. The coupling constant extends the electromagnetic vertex-counting argument to QCD by analogy. The sensitivity $S_{\text{Th}}^{\alpha_s} \sim 10^4$ is from Flambaum's 2006 estimate via dimensional transmutation; modern nuclear structure calculations have not verified this number. The prediction could be wrong by factors of several in either direction.

Why this is the primary channel. Despite the uncertainty, the strong-sector prediction is: (i) unconstrained by the PTB E3/E2 bound (Sec. 4.2); (ii) qualitatively required by DFD's gauge-coupling framework (the same logic that gives $k_\alpha = \alpha^2/(2\pi)$ gives $k_s = \alpha_s^2/(2\pi)$); (iii) amplified by the nuclear near-cancellation that guarantees $S_{\text{Th}}^{\alpha_s} \gg 1$; (iv) at a level ($\sim 8$ kHz) detectable with current per-scan precision (~ 2–4 kHz) at JILA [5,6].

7.2 Composition coupling: the lower bound

From the "family + clock" model (Appendix T of [1]), the composition difference between Th (actinide) and Sr (alkaline earth) gives:

$$K_{\text{Th}} - K_{\text{Sr}} \sim 8 \times 10^{-5}, \tag{21}$$

yielding an annual modulation:

$$\left(\frac{\delta R}{R}\right)_{\text{family}} \sim 8 \times 10^{-5} \times 1.65 \times 10^{-10} \sim 1.3 \times 10^{-14}. \tag{22}$$

This is model-dependent (the family charge assignments are phenomenological) but provides a floor: even if the strong-sector coupling is weaker than estimated, the composition channel produces a nonzero signal for any cross-species comparison.

7.3 α-sector: constrained by PTB

For completeness, the α-sector contribution using the screened coupling and Beeks's sensitivity [4]:

$$\left(\frac{\delta R}{R}\right)_\alpha = k_\alpha^{\text{eff}} \times S_{\text{Th}}^\alpha \times \delta\psi_{\text{annual}}. \tag{23}$$

However, the PTB E3/E2 bound constrains this channel. Applying the same formula to the E3/E2 ratio: $k_\alpha^{\text{eff}} \times (-6.95) \times 1.65 \times 10^{-10} = 6.9 \times 10^{-16}$, which is $53\times$ larger than the PTB measurement uncertainty of 1.3×10^{-17}.

If the coupling mechanism were purely through $k_\alpha^{\text{eff}} \times S_A^\alpha$, the PTB result would exclude $k_\alpha^{\text{eff}} = 5.98 \times 10^{-7}$ at high significance. This is *not* a problem for DFD, because the composition-dependent model (Eq. 15) predicts a null for same-ion comparisons while allowing nonzero signals for cross-species comparisons. But it does mean the α-sector cannot be treated as an independent coupling channel. The Beeks sensitivity $S_{\text{Th}}^\alpha = 5900$ quantifies the nuclear transition's response to α variation, but the *mechanism* producing $\delta\alpha/\alpha$ in a gravitational field is not pure α-coupling—it is mediated through the strong sector and composition terms.

Practical consequence. The 5.8×10^{-13} number from $k_\alpha^{\text{eff}} \times S_{\text{Th}}^\alpha \times \delta\psi$ should not be treated as an independent prediction. The strong-sector estimate (4×10^{-12}) and the composition lower bound (1.3×10^{-14}) bracket the physical prediction.

7.4 Total signal and detectability

Table 1 collects the predictions by coupling channel. The strong-sector estimate for Th-229/Sr ($\sim 4 \times 10^{-12}$, or $\sim$ 8 kHz on line b) is the primary target. The composition-coupling lower bound ($\sim 1.3 \times 10^{-14}$, or $\sim$ 26 Hz) sets a floor accessible to future nuclear clock technology. The E3/E2 null is a successful prediction of the composition-dependent model.

Current Th-229 clock performance: the JILA group has measured the nuclear transition frequency to $\sim 10^{-12}$ relative precision [5], with rapid improvements ongoing. Frequency reproducibility at $\sim 10^{-13}$ (220 Hz) has been demonstrated over 7 months in solid-state platforms [6]. A dedicated Th-229/Sr comparison campaign over one year could detect the strong-sector signal at $> 5\sigma$ with current per-scan precision ($\sim$ 2–4 kHz).

Table 1: DFD predictions for annual modulation in clock ratios. Strong-sector predictions use $k_s^{\text{eff}} = 2.4 \times 10^{-6}$; composition predictions use the family model of Appendix T [1]. $\delta\psi_{\text{annual}} = 1.65 \times 10^{-10}$. The α-sector column shows what $k_\alpha^{\text{eff}} \times \Delta S^\alpha$ would give, but this channel is constrained by PTB to $|k_\alpha| < 1.4 \times 10^{-9}$ and is not the active mechanism for any listed pair.

Channel	Composition	Strong	Status
Yb/Sr	$\sim 10^{-17}$	—	BACON: marginal
Al^+/Yb	$\sim 10^{-17}$	—	BACON: marginal
Yb^+(E3)/Sr	$\sim 10^{-16}$	—	Detectable
Hg^+/Sr	$\sim 10^{-16}$	—	Detectable
Cs/Sr	$\sim 10^{-16}$	—	Near threshold
Th-229/Sr	$\sim 10^{-14}$	$\sim 4 \times 10^{-12}$	**Decisive**
E3/E2 (same-ion)	0	0	Null (confirmed)

7.5 Falsification criteria

DFD's Th-229 predictions are sharply falsifiable:

1. **Null at 10^{-12} precision:** $|A| < 2$ kHz over 1 year $\Rightarrow$ strong-sector coupling excluded at $> 4\sigma$. This would require $|k_s^{\text{eff}} \times S_{\text{Th}}^{\alpha_s}| < 10^{-2}$, ruling out the gauge-coupling analogy at the confinement scale.

2. **Null at 10^{-14} precision:** $|A| < 0.03$ Hz $\Rightarrow$ composition coupling excluded. This would require the family charges to be effectively zero for Th/Sr, which would be in tension with existing evidence from atomic clock comparisons.

3. **Wrong phase:** If modulation is present but does not track solar orbital phase (perihelion = January 3), the mechanism is wrong.

4. **Crystal-dependent:** Different ^{229}Th:CaF_2 specimens showing different fractional amplitudes $\Rightarrow$ systematic, not nuclear.

5. **What falsifies the overall framework:** A high-precision null in Th-229/Sr *combined with* confirmed nulls in Cs/Sr, Hg/Sr, and Yb^+/Sr at 10^{-5} level would eliminate all DFD coupling channels.

8 Revised Experimental Strategy

The screening-regime constraint from BACON, combined with the coupling-channel structure from PTB E3/E2, reshapes the experimental priority list:

1. **Th-229/Sr annual modulation** (Priority 1): Largest predicted signal ($\sim 4 \times 10^{-12}$ strong-sector, $\sim 10^{-14}$ composition floor). Strong-sector estimate detectable at current per-scan precision ($\sim$ 2–4 kHz). Requires ~ 12 monthly measurements over 1 year. Timeline: achievable now with existing JILA infrastructure.

2. **Cross-species ion–neutral comparisons** (Priority 2): Yb^+(E3)/Sr and Hg^+/Sr, with predicted composition signals at $\sim 10^{-16}$. Detectable with $\sim 10^{-18}$ clocks running continuously for 1–2 years. These test the composition channel independently of the strong sector.

3. **BACON-style multi-species** (Priority 3): Predicted Yb/Sr signal at $\sim 10^{-17}$, near current BACON scatter. Would benefit from dedicated year-long campaign with improved systematics.

4. **Height-separated cavity–atom** (Priority 4, downgraded): Signal $\sim 10^{-21}$ for $\Delta h = 100$ m. Requires space-based platforms or transformative precision improvements.

9 Erratum for Section 10 of DFD v3.1

The following corrections apply to Section 10 ("Cavity-Atom Redshift Tests") of the DFD unified review, version 3.1:

9.1 What changes

1. Equation (54): $\xi_{\rm LPI}^{\rm DFD} \approx$ 1–2 is replaced by:

$$\xi_{\rm LPI}^{\rm DFD} = k_\alpha^{\rm eff}(a_{\rm local}) \cdot (S_A^\alpha - S_{\rm cav}^\alpha). \tag{24}$$

   At Earth's surface: $\xi_{\rm LPI} \sim 10^{-7}$ for typical atomic transitions.

2. The "binary test" language ("measuring $\xi_{\rm LPI} \neq 0$ at $> 5\sigma$ would falsify GR") is revised: the cavity–atom test remains a valid DFD discriminator but requires $\sim 10^{-7}$ sensitivity, not $\sim 10^{-1}$.

3. The height-separated protocol sensitivity reach changes from $|K_C - K_A| \gtrsim 10^{-5}$ (feasible) to $\sim 10^{-7}$ (requires space).

4. Nuclear clocks are added as the primary detection channel, with Th-229/Sr annual modulation of $\sim 4 \times 10^{-12}$ (strong-sector) or $\sim 10^{-14}$ (composition floor).

5. The Section 9 summary box headline $K_A = k_\alpha^{\rm eff}(a) \cdot S_A^\alpha$ should be understood as an effective parameterization for cross-species atomic comparisons, where composition differences correlate empirically with α-sensitivity differences. The fundamental coupling is through the full expression Eq. (15), with the simplified formula emerging as an approximation. For same-ion comparisons (E3/E2), the composition terms cancel, correctly predicting a null. For nuclear clocks, the dominant coupling is through the strong sector (Sec. 8b of [1]), not the electromagnetic sector.

9.2 What stays the same

1. The geometric cancellation proof (Sec. 2) is a new result that *strengthens* the framework.

2. The sector-resolved parameterization (§10.2) is unchanged.

3. The 4→3 GLS protocol (§10.3) remains valid for future multi-species tests.

4. The dispersion control analysis (§10.4) is unchanged.

5. All cross-species atom–atom predictions (Sec. 9) are preserved, with the interpretation shifting from pure α-coupling to composition-dependent coupling that correlates with α-sensitivity.

10 Conclusions

The optical metric of DFD uniquely determines constitutive relations $\varepsilon = \varepsilon_0 e^{\psi}$, $\mu = \mu_0 e^{\psi}$ through the Tamm–Plebanski formalism. These relations modify the Coulomb potential, causing electromagnetic cavity spacers to expand by $e^{+\psi}$ while light slows by $e^{-\psi}$. The compound effect gives $f_{\rm cav} \propto e^{-2\psi}$, identical to the atomic frequency scaling, so the ratio is ψ-independent at tree level.

This geometric cancellation is confirmed by three independent empirical checks and is consistent with all existing optical clock data, including the high-precision BACON three-species network. The BACON data further constrain the screening regime to be determined by the local gravitational environment (Earth's surface), not the source field (solar orbit).

The PTB Yb^+ E3/E2 null [7] provides a crucial additional constraint: pure α-sector coupling is bounded to $|k_\alpha| < 1.4 \times 10^{-9}$, confirming that the dominant coupling mechanism for cross-species comparisons is composition-dependent (through nuclear and electronic family charges), not purely electromagnetic. This is consistent with the DFD framework and explains the pattern of hints in cross-species comparisons alongside nulls in same-ion comparisons.

The physical LPI violation survives as a one-loop quantum correction, screened to $k_\alpha^{\rm eff} \sim 6 \times 10^{-7}$ at Earth's surface. For atomic clock transitions with $|S^\alpha| \lesssim 6$, the composition-dependent coupling produces annual signals at 10^{-16}–10^{-17}, at or below current sensitivity.

The transformative result is for nuclear clocks. The ^{229}Th nuclear isomer accesses the strong-sector coupling channel—invisible to atomic clocks and unconstrained by the PTB E3/E2 bound. The strong-sector estimate gives $\delta R/R \sim 4 \times 10^{-12}$ ($\sim$ 8 kHz), detectable at current per-scan precision. The composition-dependent lower bound gives $\sim 10^{-14}$ ($\sim$ 26 Hz).

The Th-229 nuclear clock is not merely a better clock for testing scalar-field coupling. It is a *qualitatively different* probe that opens a coupling channel atomic clocks cannot access. A dedicated Th-229/Sr frequency comparison over one annual cycle could provide a definitive test of DFD's predictions.

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